HEIKO HOPPE

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EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY – *Visiting Researcher, advisor: Cathy Wu* 07/26 - present

TECHNICAL UNIVERSITY OF MUNICH 07/24 - present
Ph.D., Reinforcement Learning and Operations Research, advisor: Maximilian Schiffer
- First-authored 3 papers in 2 years (2x NeurIPS, 1x L4DC)
- Invited talks at ICSP, MIT, University of Twente; reviewer for ICML and EJOR

POLYTECHNIQUE MONTRÉAL – *Visiting Researcher, advisor: Quentin Cappart* 07/23 - 10/23

TECHNICAL UNIVERSITY OF MUNICH
M.Sc. Finance and Information Management (Grade: 4.0 / 4.0, top 5%) 10/21 - 12/23
B.Sc. Management and Technology (Grade: 3.7 / 4.0, top 5%) 10/18 - 09/21

EXPERIENCE

TECHNICAL UNIVERSITY OF MUNICH 07/24 - present
Research Scientist
- Developed machine learning algorithm for supply chain optimization for software company; $>$80% solution time reduction
- Built reinforcement learning algorithm for production scheduling; 50% improvement over baseline
- Taught *Intro. to DRL* (lecture, 400 part.), *Intro. to Python* (seminar, 20 part.); supervised 6 master theses and 10 student projects

BOSTON CONSULTING GROUP 04/24 - 06/24
Consulting Intern
- Received 5.0 / 5.0 evaluation and full-time return offer

INFINEON TECHNOLOGIES 01/24 - 03/24
Data Science Intern
- Developed explainable revenue forecasting algorithm; explained $>$90% of forecast variance

SELECTED PUBLICATIONS

Breaking the Grid: Distance-Guided Reinforcement Learning in Large Discrete Action Spaces NeurIPS 2026
With Fabian Akkerman (Univ. Twente), Wouter van Heeswijk (Univ. Twente), and Maximilian Schiffer (TUM)
Developed RL algorithm for large discrete action spaces (10^{20} actions), using latent action mappings to improve performance by 66%

Structured Reinforcement Learning for Combinatorial Decision-Making NeurIPS 2025
With Léo Baty (ENPC), Louis Bouvier (ENPC), Axel Parmentier (ENPC), and Maximilian Schiffer (TUM)
Proposed an RL algorithm for combinatorial action spaces, leveraging Fenchel-Young Losses to improve performance by 92%

Global Rewards in Multi-Agent Deep Reinforcment Learning for Autonomous Mobility on Demand Systems L4DC 2024
With Tobias Enders (TUM), Quentin Cappart (MILA), and Maximilian Schiffer (TUM)
Invented RL algorithm combining SAC with counterfactual Q-values to align global rewards with individual incentives of 2000 agents

Combinatorial Optimization Augmented Machine Learning Received revision at EJOR
With Maximilian Schiffer (TUM), Yue Su (Univ. Lille), Louis Bouvier (ENPC), and Axel Parmentier (ENPC)
Conceptualized methodological guide and research outlook on algorithms combining machine learning and combinatorial optimization

SKILLS AND SCHOLARSHIPS

Programming	Python (Pytorch, Tensorflow, Pandas, Numpy), Julia, R, SQL
Tools & Infrastructure	Git, Slurm (HPC Clusters), Linux/Bash, Docker
Scholarships	Linde/MDSI Ph.D. Fellowship, Max Weber Program (part of Studienstiftung, $<$10% acceptance rate)
Extracurricular activities	Integreat (supported international scaling of refugee integration app), Because We Care (trained $>$100 high school students in financial literacy), Academy Consult (led finance team in student consulting club)